llmXive Automated Review of Mega-ASR: Towards In-the-wild^2 Speech Recognition via Scaling up Real-world Acoustic Simulation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several specific factual claims regarding dataset composition, hyperparameter settings, and the justification for methodological choices that require verification against the provided text and tables.

First, there is a direct contradiction regarding the WER-gated threshold (τ) used in the Dual-Granularity WER-Gated Policy Optimization (DG-WGPO). The main text in Section 4.2 explicitly states: "We set the three hyperparameters as $\tau = 0.3$..." However, Table 3 in the Appendix (labeled `tab:dg_wgpo_reward_hparams`) lists the "WER gate threshold τ" as 0.5. Since the ablation study in Table 2 (`tab:hp-tau`) shows that $\tau = 0.3$ yields the best result (7.64 WER) compared to 0.2, 0.4, and 0.5, it is likely the text is correct and the table is a typo. However, this discrepancy must be resolved to ensure the reproducibility of the reported results.

Second, the count of acoustic phenomena is inconsistent. The Abstract and Section 3 state that the `Voices-in-the-wild-2M` dataset covers 7 classic acoustic phenomena. However, Table 1 in the Appendix (`tab:primitive_effects`) lists 8 primitive effects (Additive noise, Echo delay, Reverberation, Nonlinear distortion, Resampling, Spectral filtering, Loudness transformation, Frame-level stutter). While the text mentions these are mapped to 7 "atomic" effects, the transition from 8 primitives to 7 atoms is not explicitly detailed in the main text, creating ambiguity about which primitive was merged or excluded. The claim of "7 phenomena" should be clarified to align with the 8 primitives listed in the appendix.

Third, the justification for the 30% WER threshold relies on the claim that error modes shift "abruptly" beyond this point. While the authors propose a gated reward to handle this, the sensitivity analysis in Table 2 (`tab:hp-tau`) shows that varying τ between 0.2 and 0.4 results in very small WER differences (7.68 to 7.64). The term "abrupt" suggests a sharp discontinuity that the provided data does not strongly support. The claim should be moderated to reflect that the threshold was empirically optimized rather than being a distinct, sharp boundary in error behavior.

Finally, the claim in the Abstract that the dataset includes "54 physically plausible compound scenarios" is supported by the enumeration in the Appendix (`tab:compound_scenario_count`), which sums to 54. This claim is accurate. The citation of external datasets (e.g., MUSAN, DNS) and the specific WER numbers in the main tables appear consistent with the provided text, though the specific values in the "Case Study" (e.g., 100.0% WER for Qwen3-ASR) are presented as qualitative examples and are not contradicted by the tables.
Action items.

- [writing] In Section 4.2 (DG-WGPO), the text states the WER-gated threshold is set to []=0.3, but Table 3 (tab:dg_wgpo_reward_hparams) in the Appendix lists []=0.5. This contradiction must be resolved to ensure the reported hyperparameters match the experimental setup.

- [writing] The abstract and Section 3 claim the dataset covers '7 classic acoustic phenomena,' yet Table 1 (tab:primitive_effects) in the Appendix lists '8 primitive effects' before aggregation. Clarify whether one effect was merged or excluded to ensure the count of 7 is accurate and consistent with the simulation pipeline.
- [writing] Section 4.2 claims that errors shift 'abruptly' at WER > 30%, justifying the gated reward. However, the ablation study in Table 2 (tab:hp-tau) shows only a marginal WER difference (7.68 vs 7.64) between □=0.2 and □=0.3. The text should temper the claim of an 'abrupt' shift or provide stronger evidence that the 30% threshold is the critical inflection point.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure presents a radar chart comparison, but the radial axis scaling on the left plot is misleading, causing low numerical scores to appear visually high. Additionally, the legend text is partially cropped at the bottom edge.

Figure 2

Figure 2 is a clear and effective infographic that visually communicates the scale and diversity of the 'Voices-in-the-wild-2M' dataset. The text overlay explicitly lists the 7 domains and 54 hybrid scenarios mentioned in the caption, while the background collage and network diagram illustrate the variety of acoustic degradations.

Figure 3

Figure 3 presents training curves and a sampling distribution comparison, but the right panel's unlabeled y-axis and undefined legend terms undermine clarity, and the left panel's y-axis labeling is inconsistent across subplots.

Figure 4

The figure is a complex workflow diagram but is rendered useless by a placeholder caption that fails to describe the content or define the numerous symbols and abbreviations used.

Figure 5

The figure provides a clear visual workflow for the reward mechanism but fails to visually represent the 'DG-WGPO' framework named in the caption, and the mathematical formula within the Word-level Reward box is too blurry to read.

Figure 6

The figure is a detailed pipeline diagram, but the caption is a placeholder ('Enter Caption'), which is a critical error that prevents the figure from being understood or verified.

Figure 7

The figure effectively demonstrates the qualitative advantages of Mega-ASR in challenging scenarios. However, the quantitative metrics (WER) contain inconsistencies, specifically the '–' label for Ground Truth and a potential mismatch between the 14.3% WER and the identical text output in the Entity Recovery case.

Figure 8

Figure 8 is a clear and well-constructed line plot that effectively visualizes the four difficulty mapping functions described in the caption. All axes are properly labeled with units, the legend is distinct, and the curves accurately reflect the textual descriptions of how each function transforms the uniform sample into the severity variable.

Figure 9

Figure 9 effectively presents five qualitative case studies comparing Qwen3-ASR and Mega-ASR across distinct error modes. The layout is clear, with reference transcripts, model outputs, and specific observations for each example, fully supporting the caption's claims. Action items.

- [science] Figure 1: The left radar chart's radial axis is labeled '100' at the outermost ring, but the data points for 'CommonVoice22' (6.97) and 'Voxpopuli' (6.25) are plotted near the 80% mark, implying the axis represents a percentage scale (0-100%) while the labels suggest absolute values or a different metric. This creates a severe visual distortion where low scores appear high.
- [writing] Figure 1: The legend at the bottom is cut off on the left side, making the full name of the blue line ('Qwen3-ASR-1.7B(Previous SOTA)') partially illegible.
- [science] Figure 3: The right panel's y-axis (0.45–0.65) is unlabeled, making it impossible

to interpret the ‘comparison of difficulty sampling distributions’ claimed in the caption.

- [science] Figure 3: The right panel legend lists four sampling methods (Gaussian, Linear, Sqrt-Bwd, Sqrt-Fwd), but the caption does not define these terms or link them to the left panel’s ‘atomic effects’.
- [writing] Figure 3: Left panel subplots lack individual y-axis labels; while ‘1 - WER’ is shown on the first subplot, it is not repeated or clarified for the other seven, risking misinterpretation.
- [fatal] Figure 4: The caption is a placeholder (‘Enter Caption [figure3.pdf]’) and does not describe the figure content, making it impossible to verify if the figure supports its claims.
- [fatal] Figure 4: The filename in the caption (‘figure3.pdf’) contradicts the figure label (‘Figure 4’), indicating a likely copy-paste error or mislabeling.
- [science] Figure 4: The diagram contains undefined abbreviations and symbols (e.g., ‘q’, ‘O1’, ‘R1’, ‘A1’, ‘WER’, ‘LCS’) without a legend or explanation in the caption.
- [science] Figure 5: The caption describes the framework as ‘DG-WGPO’, but the diagram explicitly labels the initialization stage as ‘A2S-SFT’ and the core mechanism as ‘Recov-Recon Dynamic Reward’ without ever mentioning ‘DG-WGPO’ or ‘gated fusion’ in the visual elements, creating a disconnect between the text and the figure content.
- [writing] Figure 5: The ‘Word-level Reward’ box contains a formula for ‘n_recal’ that is illegible due to low resolution; the subscripts and specific terms in the fraction cannot be read.
- [fatal] Figure 6: The caption is ‘Enter Caption [figure3.pdf]’, indicating a placeholder error where the figure content (likely a pipeline diagram) does not match the assigned caption file or text.
- [science] Figure 6: The diagram displays a complex ‘Recov-Recon Dynamic Reward’ framework with specific components like ‘A2S-SFT’ and ‘Mega-ASR’, but the missing caption fails to define these terms or explain the workflow, rendering the figure unintelligible without external context.
- [science] Figure 7: The ‘WER’ (Word Error Rate) for the Ground Truth row is labeled as ‘–‘, which is technically incorrect as the WER of a reference against itself is 0.0%. This creates a confusing baseline for the comparison.
- [writing] Figure 7: The WER for the ‘Qwen3-ASR’ model in the ‘Far field’ case is listed as 100.0% for an ‘<Empty>’ output. While understandable, WER is typically undefined or requires a specific convention for empty predictions; a note or alternative metric (like ‘Empty Output’) would be clearer.
- [writing] Figure 7: The ‘Entity Recovery’ case shows a WER of 14.3% for Mega-ASR, yet the text output appears to match the Ground Truth exactly (ignoring capitalization). If

the WER is non-zero, the text should reflect the specific error (e.g., 'VictorNet' vs 'Victor Company'), or the WER should be 0.0%.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology and acronyms that are not defined for a broader audience, creating a barrier to entry for non-specialist readers.

First, several acronyms are introduced without expansion. "LALMs" (Large Audio-Language Models) appears in the Introduction without being spelled out. "WER" (Word Error Rate) is used frequently in the Abstract and Introduction but is never explicitly defined. "LoRA" (Low-Rank Adaptation) is mentioned in the context of the router and training details without definition. "RIR" (Room Impulse Response) is used in the Related Work section without explanation. These should be defined at their first occurrence.

Second, the paper uses unnecessary jargon where plain English would suffice. The term "atomic acoustic effects" is used repeatedly to describe basic distortions; "basic" or "fundamental" would be clearer. "Agentic check" in the dataset construction section is vague; "automated validation" is more precise. "Backbone preservation" in the reward function description is metaphorical; "structural integrity" is more standard. "Rollouts" is a specific reinforcement learning term that should be replaced with "generated hypotheses" or "candidate outputs" for clarity.

Finally, the phrase "in-the-wild^2" in the title and throughout the text is a stylistic choice that may confuse readers. While it attempts to convey a second-order complexity, it is not standard terminology and should be explained or rephrased (e.g., "extremely complex real-world conditions").

Addressing these issues will significantly improve the paper's accessibility without compromising its technical rigor.

Action items.

- [writing] The manuscript relies heavily on specialized terminology and acronyms that are not defined for a broader audience, creating a barrier to entry for non-specialist readers. First, several acronyms are introduced without expansion. "LALMs" (Large Audio-Language Models) appears in the Introduction without being spelled out. "WER" (Word Error Rate) is used frequently in the Abstract and Introduction but is never explicitly defined. "LoRA" (Low-Rank Adaptation) is mentioned in the context of the route

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent logical framework for addressing the "acoustic robustness bottleneck" by linking the failure modes of existing models (semantic collapse at high WER) to a specific architectural solution (Dual-Granularity WER-Gated Policy Optimization). The premise that standard WER rewards fail to provide gradients when WER > 30% is well-motivated by the observed shift from word-level errors to hallucinations. The proposed mechanism of switching reward granularity based on a WER threshold logically follows from this premise.

However, the internal consistency of the experimental reporting is compromised by significant contradictions in key hyperparameters, which prevents the verification of the causal claims made in the ablation studies.

First, there is a direct contradiction regarding the WER-gated threshold (τ) used in the DG-WGPO reward function. In Section 4.2.2 ("WER-gated dynamic fusion"), the text explicitly states: "We set the three hyperparameters as $\tau = 0.3$..." and the fusion logic is defined around this value. However, in the Appendix, under the table "Reward hyperparameters used in DG-WGPO" (Table 4) and the accompanying text in "Reward tuning and diagnostics," the value is listed as $\tau = 0.5$. Since the ablation study in Table 2 (Sensitivity to gating threshold τ) shows that $\tau = 0.3$ yields the best result (7.64 WER) compared to $\tau = 0.5$ (7.70 WER), the discrepancy between the main text's claim of using the optimal value and the appendix's reporting of a sub-optimal value (or vice versa) creates a logical gap. It is unclear which threshold was actually used to generate the "Mega-ASR (full)" results in Table 1.

Second, the number of rollouts (K) used during the RL training phase is inconsistent. Section 5.1 ("Implementation Details") states: "RL runs for 6,000 steps with... $K = 16$ rollouts per input." Conversely, Appendix Table 3 ("Generation settings used in the main DG-WGPO run") lists "Number of generations" as 12. Furthermore, the "Effective prompt batch size" calculation in that same table is shown as $4 \times 3 \times 16 = 192$, which mathematically relies on $K = 16$, yet the table explicitly lists 12 generations. This inconsistency casts doubt on the reproducibility of the training dynamics and the validity of the advantage estimation described in the methodology.

Finally, while the logic of the "Environment-Aware Routing" is sound (preserving clean-speech performance), the specific claim in Section 5.1 that routing "improves LibriSpeech WER from 1.78/3.57 to 1.63/3.37" requires scrutiny. The baseline Qwen3-ASR (Table 2) is listed as 1.62/3.07 (Dev) and 1.62/3.40 (Test). The "Ours w/ router" row shows 1.64/3.07 (Dev) and 1.63/3.37 (Test). The text claims an improvement from 1.78 (which matches the "Ours" without router Dev set), but the comparison baseline in the text (1.78) does not match the baseline model's reported score in the table (1.62). This suggests a potential confusion between the base model's performance and the non-routed model's performance in the narrative, weakening the logical support for the routing claim.

These inconsistencies do not necessarily invalidate the core scientific contribution but require clarification to ensure the reported results logically follow from the described experimental setup.

Action items.

- [science] Inconsistent WER gate threshold: Section 4.2.2 defines the WER-gated fusion threshold as []=0.3, but Appendix Table 4 (Reward hyperparameters) and the text in Appendix Section 'Reward tuning and diagnostics' state []=0.5. This contradiction undermines the reproducibility of the DG-WGPO mechanism.
- [science] Contradictory generation counts: Section 5.1 'Implementation Details' states K=16 rollouts per input, while Appendix Table 3 (DG-WGPO generation hyperparameters) lists 'Number of generations' as 12. The effective batch size calculation in the same table (4x3x16=192) also conflicts with the stated 12 generations.
- [writing] Inconsistent hyperparameter reporting: Section 4.2.2 sets []_dyn=0.6 and []_s=0.4, which matches Table 4, but the text in Appendix 'Reward tuning and diagnostics' repeats these values while the table caption for Table 4 lists 'Low-WER fusion' as 0.75/0.25 and 'High-WER fusion' as 0.25/0.75, which is consistent, but the text in Section 4.2.2 does not explicitly define the fusion weights, only the threshold logic, creating ambiguity in the exact formula application.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The review focuses on potential over-claiming and extrapolation beyond the provided evidence.

1. Generalization of Performance Metrics: The abstract and conclusion repeatedly claim "over 30% relative WER reduction" against strong baselines. While Table 3 (Voices-in-the-Wild-Bench) supports this for specific mixed degradation scenarios (e.g., 65.8% reduction on mixed degradations), the text generalizes this finding to "complex compositional acoustic scenarios" broadly. However, Table 1 (NOIZEUS, CHiME-4, VOiCES) shows relative reductions ranging from ~15% to ~17.4% (e.g., 19.80 vs 23.97 on NOIZEUS 0dB is ~17.4%). The claim of "over 30%" is not supported by the aggregate results on standard adverse-condition benchmarks, creating a misleading impression of the model's general robustness gains. The text should qualify this claim to apply specifically to the *Voices-in-the-Wild-Bench* mixed scenarios or provide a weighted average that justifies the 30% figure.

2. Unsubstantiated "Physical Plausibility" Verification: In Section 3, the authors state that

the 54 compound scenarios were generated with an "agentic check that verifies physical plausibility." The paper provides no details on the agent's architecture, the rules it enforces, or the validation process (e.g., did it reject any combinations?). Without describing the mechanism or reporting the rejection rate, the claim that all 54 scenarios are "physically plausible" is an unsupported assertion. The term "physically plausible" implies a rigorous simulation of acoustic physics, which is not evidenced by the description of simple spectral manipulation and composition.

3. "Human-Level Performance" Claim: The introduction claims that large audio-language models achieve "human-level performance on canonical benchmarks." While the reported WERs (e.g., 1.63% on LibriSpeech) are impressive, the term "human-level" is a specific benchmark in speech recognition that usually refers to the error rate of human transcribers on the same test set. The paper does not cite a human baseline WER for the specific datasets used (e.g., LibriSpeech test-clean) to substantiate this comparison. Given that human error on clean speech is often cited as <1% or ~0.5% in ideal conditions, claiming "human-level" based solely on model WER without a direct human comparison is an over-extrapolation.

4. Scope of "In-the-wild" vs. Simulation: The title and abstract emphasize "In-the-wild" recognition, yet the primary dataset (Voices-in-the-wild-2M) is entirely synthetic/simulated. While the authors argue this is a scalable paradigm, the leap from "simulated compound scenarios" to "real-world in-the-wild" robustness is a significant extrapolation. The paper relies on the "Voices-in-the-wild-Bench" (which contains 1,500 real recordings) for validation, but the training data is 100% synthetic. The claim that the model establishes a "scalable paradigm for robust ASR in-the-wild" based primarily on synthetic training data requires stronger caveats regarding the domain gap between simulation and reality, which are currently understated.

Action items.

- [writing] The claim of 'over 30% relative WER reduction' in the abstract and conclusion is not universally supported by the provided tables. Table 1 shows a 17.4% reduction on NOIZEUS 0dB and ~15-20% on other subsets. The '30%' figure likely applies only to specific compound scenarios in Voices-in-the-Wild-Bench (Table 3), but the text generalizes this to 'complex compositional acoustic scenarios' without explicit qualification, risking over-claiming on general benchmarks.
- [science] The paper claims the dataset covers '54 physically plausible compound scenarios' verified by an 'agentic check' (Section 3). However, the methodology for this 'agentic check' is not described, nor is the failure rate or criteria for 'physical plausibility' defined. Without this evidence, the claim of physical plausibility for all 54 scenarios is an unsupported extrapolation.
- [writing] The abstract states the model achieves 'human-level performance' on canonical benchmarks, yet Table 2 shows WERs of 1.63-3.57 on LibriSpeech. While low, 'human-level' is a strong claim that typically requires comparison to human transcription error rates (often cited as ~5% for difficult speech, but <1% for clean). The paper does not pro-

vide this specific human baseline comparison to justify the 'human-level' terminology.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a significant technical contribution to robust ASR but raises several safety and ethics concerns that require clarification before acceptance.

First, regarding human subjects, Section 3.3 ("Voices-in-the-wild-Bench") explicitly mentions the collection of "1,500 real-world recordings... from 16 human participants." However, the manuscript contains no mention of Institutional Review Board (IRB) approval, ethical oversight, or the specific informed consent process used for these participants. Given that these recordings are part of a public benchmark, the authors must provide evidence of ethical compliance, including how participant privacy was protected and whether consent covered the specific use of their voice data for training and evaluating AI models. Without this, the inclusion of human data is a fatal ethical oversight.

Second, the dataset construction (Section 3.2) aggregates clean speech from multiple public sources (LibriSpeech, Common Voice, etc.) to create a new, large-scale synthetic dataset. The authors must explicitly address the licensing compatibility of these source materials. Specifically, they need to confirm that the terms of service and licenses of the source datasets permit the creation of a derivative work and its subsequent public release. Ambiguity here could lead to copyright infringement or license violation, particularly if the source data has restrictions on commercial use or redistribution.

Finally, while the paper focuses on improving robustness, the techniques described—specifically the simulation of "electronic distortion" and "transmission dropout" to create adversarial-like conditions—carry potential dual-use risks. The authors should include a brief discussion in the "Limitations" or "Ethical Considerations" section acknowledging that these simulation methods could theoretically be repurposed to generate adversarial audio examples designed to bypass ASR security filters or create undetectable deepfakes. A proactive statement on the intended defensive use of this technology is recommended.

Action items.

- [fatal] The paper states the evaluation benchmark includes '1,500 real-world recordings collected from internet sources and 16 human participants' (Section 3.3) but lacks explicit details on IRB approval, informed consent procedures, or data privacy safeguards for these participants. This is a critical ethical gap for human-subject research.
- [science] The dataset construction relies on 'clean speech from LibriSpeech, Common Voice, WenetSpeech, and AISHELL-1' (Section 3.2). The authors must explicitly confirm that the licensing terms of these source datasets permit the creation of a new, deriva-

tive synthetic dataset ('Voices-in-the-Wild-2M') and its public release, including any restrictions on commercial use or redistribution.

- [writing] The methodology involves generating '54 physically plausible compound scenarios' including 'electronic distortion' and 'transmission dropout' (Section 3.2). While intended for robustness, the authors should briefly discuss potential dual-use risks, such as the possibility of these simulation techniques being adapted to generate adversarial audio attacks or deepfake audio that evades detection systems.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence presented in "Mega-ASR" is generally strong, supported by extensive ablation studies and comparisons against a wide range of baselines. The proposed dataset, "Voices-in-the-wild-2M," addresses a clear gap in compositional acoustic scenarios, and the dual-granularity reward mechanism (DG-WGPO) is logically motivated by the observed shift in error modes at high WER. The ablation results in Table 4 effectively isolate the contribution of each component, particularly highlighting the necessity of the sentence-level structural reward for handling hallucinations.

However, several aspects of the evidence require strengthening to fully support the central claims of robustness and generalization. First, while the paper reports significant relative WER reductions (e.g., "over 30%"), it lacks statistical significance testing. Word Error Rate (WER) can exhibit high variance across different acoustic conditions and utterance lengths. Without confidence intervals or results from statistical tests (such as bootstrap resampling or paired t-tests) on the benchmark results, it is difficult to ascertain whether the observed improvements are statistically significant or potentially due to random variation in the test set.

Second, the primary evaluation relies heavily on the synthetic "Voices-in-the-wild-2M" dataset and its derived benchmark. While the authors validate the simulator against real data during the construction phase, the final performance claims are largely based on this synthetic environment. The paper would benefit from a more rigorous evaluation on a completely independent, held-out real-world dataset that was not involved in any stage of the simulation calibration or model training. This would provide stronger evidence that the learned robustness generalizes to truly "in-the-wild" conditions rather than overfitting to the specific spectral artifacts of the simulation pipeline.

Finally, the sensitivity of the WER-gated fusion threshold (τ) warrants further investigation. The ablation study in Table 6 shows that $\tau = 0.3$ yields the best results, but the performance drop-off for $\tau = 0.2$ and $\tau = 0.4$ is relatively modest. A more detailed analysis of how the model's performance varies across a continuous range of τ values, or an analysis

of the distribution of WERs in the training data relative to this threshold, would strengthen the justification for this specific hyperparameter choice and demonstrate the stability of the proposed reward mechanism.
Action items.

- [science] The paper claims a 30% relative WER reduction on compositional scenarios but does not report confidence intervals or statistical significance tests (e.g., bootstrap or paired t-tests) for these gains. Given the high variance in WER across different acoustic conditions, statistical validation is required to confirm the robustness of the reported improvements.
- [science] The 'Voices-in-the-wild-2M' dataset is entirely synthetic, generated via spectral manipulation. The paper lacks a rigorous cross-validation experiment demonstrating that performance gains on this synthetic benchmark transfer to a held-out, purely real-world dataset not used in any training or validation step. Without this, the claim of 'in-the-wild' robustness is partially unverified.
- [science] The ablation study (Table 4) shows that removing the sentence-level structural reward (R_{struc}) causes the largest degradation. However, the paper does not provide a sensitivity analysis on the specific threshold $au = 0.3$ used for the WER-gated fusion. A small shift in this threshold could significantly alter the balance between token-level and sentence-level rewards, yet the robustness of this hyperparameter choice is not fully explored.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical rigor of the evaluation and ablation studies requires minor revision to fully support the claims of superiority.

First, the paper presents extensive Word Error Rate (WER) comparisons across multiple benchmarks (Tables 1, 2, 3). While the absolute differences are often substantial (e.g., 19.80% vs 23.97% on NOIZEUS 0dB), the manuscript does not report statistical significance tests (such as paired t-tests, bootstrap confidence intervals, or McNemar's test) to verify that these improvements are statistically significant rather than artifacts of random initialization or stochastic decoding. Given the high variance often observed in ASR evaluation on small subsets, providing confidence intervals for the reported WERs is essential.

Second, there is a critical inconsistency regarding the WER-gated threshold τ in the Dual-Granularity WER-Gated Policy Optimization (DG-WGPO) framework. The main text (Section 4.2, "WER-gated dynamic fusion") explicitly states: "We set the three hyperparameters as $\tau = 0.3$". However, Table 6 in the Appendix ("Reward hyperparameters used in DG-WGPO") lists "WER gate threshold τ" as 0.5. This discrepancy is significant because τ determines the switch between token-level and sentence-level reward dominance. The authors must clarify which value was used for the final reported results and ensure consistency throughout the text and tables.

Third, the ablation study in Table 3 ("Reward design") compares the rule-based reward against an LLM-judge reward. The authors claim the rule-based approach achieves "comparable WER" (differences within ~0.1) with significantly lower time cost. However, the table presents single-point estimates without any measure of variance (e.g., standard deviation over multiple random seeds or runs). Without variance estimates, it is impossible to determine if the 0.1 WER difference is statistically meaningful or within the noise floor of the training process. The authors should report results averaged over multiple seeds with standard deviations to substantiate the claim of comparable performance.

Finally, the sensitivity analysis in Table 4 and Table 5 relies on single-run WER values to select hyperparameters ($\alpha_{dyn}, \alpha_s, \tau$). While the trends are visible, the lack of error bars or confidence intervals makes it difficult to assess the robustness of the selected hyperparameters against training stochasticity. Including these statistical measures would strengthen the validity of the hyperparameter selection process.
Action items.

- [science] The paper reports specific WER values (e.g., 19.80 vs 23.97 in Table 1) but lacks statistical significance testing (e.g., paired t-tests or bootstrap confidence intervals) to confirm these improvements are not due to random variance, especially given the stochastic nature of RL training.
- [science] The WER-gated threshold τ is set to 0.3 in the main text (Section 4.2) but listed as 0.5 in Table 6 (Appendix). This inconsistency in a critical hyperparameter governing the reward fusion strategy must be resolved and justified.
- [science] The ablation study in Table 3 compares rule-based vs. LLM-judge rewards based on a single run's WER and time cost. It lacks variance estimates (e.g., standard deviation over multiple seeds) to validate the claim that performance is 'comparable' within ~0.1 WER.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling technical contribution, but the writing quality requires minor revisions to meet the standards of a top-tier publication. While the overall structure is logical, there are several instances of grammatical fragmentation, inconsistent formatting, and informal notation that disrupt the reading flow.

Specifically, in Section 3.1, the sentence "mild WER typically between\textbf{ 4\%–10\%}" suffers from a missing space before the bold command and awkward punctuation. This type of typographical error, while minor, detracts from the professional polish of the paper. Similarly, in Section 4.1, the phrase "instilling perceptual robustness and semantic recovery" appears as a dangling modifier or fragment, failing to connect grammatically to the main clause. This should be rephrased to ensure the sentence is syntactically complete.

In Section 4.2, the authors use the symbol "<=" to denote inequality (≤). In formal academic writing, mathematical symbols should be used rather than programming-style operators. Additionally, the sentence structure in this section is somewhat dense and could be simplified for clarity.

Finally, there is a lack of consistency in the capitalization of the dataset name "Voices-in-the-wild-2M" versus "Voices-in-the-Wild-2M" throughout the text and captions. Standardizing this nomenclature is essential for readability and professional presentation. Addressing these issues will significantly improve the clarity and flow of the manuscript without altering the scientific content.

Action items.

- [writing] In Section 3.1, the phrase 'between extbf{ 4\%–10\%}' contains a missing space before the bold command and inconsistent punctuation. It should be 'between 4\%–10\%' or 'between 4\% and 10\%'. Additionally, the bolding of the numbers within the sentence flow is stylistically inconsistent with the rest of the manuscript.
- [writing] Section 4.1 contains a sentence fragment: 'instilling perceptual robustness and semantic recovery.' This phrase lacks a main verb and is grammatically disconnected from the preceding clause. It should be integrated into the sentence (e.g., '...via ..., instilling...') or rewritten as a complete sentence.
- [writing] In Section 4.2, the text states 'We observe during training that errors when $extWER$<=30%...' The use of the symbol '<=' is informal for a research paper; it should be written as '≤' or 'less than or equal to'. Furthermore, the sentence structure is slightly clunky and could be smoothed for better flow.
- [writing] Throughout the manuscript, there are inconsistent capitalizations in section titles and figure captions (e.g., 'Voices-in-the-wild-2M' vs 'Voices-in-the-Wild-2M'). The dataset name should be consistently capitalized (likely 'Voices-in-the-Wild-2M') to maintain professional polish and readability.